

changes the composition of labor demand across junior and senior workers, holding firm scale fixed, rather than to model total employment in general equilibrium. A firm produces Y by combining a continuum of tasks indexed by $x \in [0, 1]$ with a CES technology:

$$Y = \left(\int_0^1 y(x)^{\frac{\sigma-1}{\sigma}} dx \right)^{\frac{\sigma}{\sigma-1}}, \quad \sigma > 0, \sigma \neq 1, \quad (1)$$

where $y(x)$ is the output of task x and σ is the elasticity of substitution across tasks (the case $\sigma = 1$ is the Cobb-Douglas limit). The firm employs junior workers L_J and senior workers L_S . Tasks are ordered by complexity, reflecting the hierarchical structure of knowledge-based organizations (Garicano, 2000; Garicano and Rossi-Hansberg, 2006). When task x is performed by a worker of type $k \in \{J, S\}$, output is

$$y(x) = \gamma_k(x) \cdot l_k(x), \quad (2)$$

where $\gamma_k(x)$ is type- k productivity and $l_k(x)$ is the labor allocated. We assume:

Assumption 1 (Comparative advantage). *The productivity ratio $\gamma_S(x)/\gamma_J(x)$ is continuously differentiable and strictly increasing in x , with $\frac{d}{dx} \ln\left(\frac{\gamma_S(x)}{\gamma_J(x)}\right) > 0$ for all $x \in (0, 1)$.*

Under cost minimization, there exists a threshold $\hat{x}$ such that juniors perform all tasks $x < \hat{x}$ and seniors perform all tasks $x \geq \hat{x}$.¹²

GenAI affects the production technology through two channels. First, *task displacement*: GenAI automates all tasks $x \in [0, \alpha]$ for some exogenous threshold $\alpha \in (0, \hat{x})$, removing them from human labor; automated tasks remain in the CES aggregator and are performed by GenAI at exogenous unit cost $c_A(x) < w_J/\gamma_J(x)$ for $x \in [0, \alpha]$.¹³ Second, *within-task augmentation*: on tasks that remain human-performed, GenAI augments productivity to $\tilde{\gamma}_k(x) = (1 + \delta_k) \cdot \gamma_k(x)$, where $\delta_J \geq \delta_S \geq 0$ captures the experimental finding that GenAI disproportionately increases the productivity of less-experienced workers.¹⁴

¹²The threshold $\hat{x}$ satisfies $\gamma_J(\hat{x})/w_J = \gamma_S(\hat{x})/w_S$, where w_J and w_S are wages.

¹³We take α as given rather than modeling the adoption decision. The framework is a model of AI's incidence across worker types, not of firms' adoption choices. The assumption $\alpha < \hat{x}$ is a simplifying benchmark reflecting the empirically relevant case in which the bulk of automated tasks are routine tasks that juniors perform (Eloundou et al., 2024). If some senior tasks are also automated, seniors face a displacement channel too, but the asymmetry remains as long as a larger share of automated tasks falls in the junior range. The condition $c_A(x) < w_J/\gamma_J(x)$ ensures that automation is cost-reducing: only tasks for which GenAI is cheaper than human labor are automated.

¹⁴The key results extend to task-specific augmentation $\delta_k(x)$ as long as the post-AI productivity ratio

3.3 Effect of GenAI on Labor Demand by Seniority

After the firm adopts GenAI, juniors perform tasks $(\alpha, \hat{x}')$ and seniors perform tasks $[\hat{x}', 1]$, where $\hat{x}'$ is the post-AI allocation threshold, determined by the standard equal-cost condition under augmented productivities (see Supplemental Appendix A.6 for the derivation):

$$\frac{\gamma_S(\hat{x}')}{\gamma_J(\hat{x}')} = \frac{1 + \delta_J}{1 + \delta_S} \cdot \frac{\gamma_S(\hat{x})}{\gamma_J(\hat{x})}. \quad (3)$$

Since we assume that GenAI augments junior productivity at least as much as senior productivity ($\delta_J \geq \delta_S$), the right-hand side weakly exceeds $\gamma_S(\hat{x})/\gamma_J(\hat{x})$, and because $\gamma_S(x)/\gamma_J(x)$ is strictly increasing (Assumption 1), it follows that $\hat{x}' \geq \hat{x}$: the junior task set expands among surviving tasks. When $\delta_J = \delta_S$, the threshold is unchanged exactly.

The change in junior labor demand can be decomposed into three components. We denote by $l_k^{*pre}(x)$ and $l_k^{*post}(x)$ the cost-minimizing labor allocations before and after AI adoption. Total junior labor demand is

$$L_J^{*pre} = \int_0^{\hat{x}} l_J^{*pre}(x) dx, \quad L_J^{*post} = \int_{\alpha}^{\hat{x}'} l_J^{*post}(x) dx, \quad (4)$$

and the change is $\Delta L_J^* \equiv L_J^{*post} - L_J^{*pre}$.

Proposition 1 (Decomposition). *The change in junior labor demand following GenAI adoption is:*

$$\Delta L_J^* = \underbrace{-\int_0^{\alpha} l_J^{*pre}(x) dx}_{\text{displacement } (<0)} + \underbrace{\int_{\alpha}^{\hat{x}} [l_J^{*post}(x) - l_J^{*pre}(x)] dx}_{\text{productivity } (\leq 0)} + \underbrace{\int_{\hat{x}}^{\hat{x}'} l_J^{*post}(x) dx}_{\text{reallocation } (\geq 0)}. \quad (5)$$

Figure 1 illustrates the task partition before and after GenAI adoption. The three components of the decomposition have distinct determinants and signs. First, the *displacement effect* is the direct loss of automated junior tasks. Its magnitude is increasing in α —the extent of automation—and it is unambiguously negative.

Second, the *productivity effect* captures the change in labor needed per surviving task. Under the partial-equilibrium assumption that aggregate output Y and wages are fixed, it

$\tilde{\gamma}_S(x)/\tilde{\gamma}_J(x)$ remains strictly monotone increasing, so that a unique threshold continues to exist.

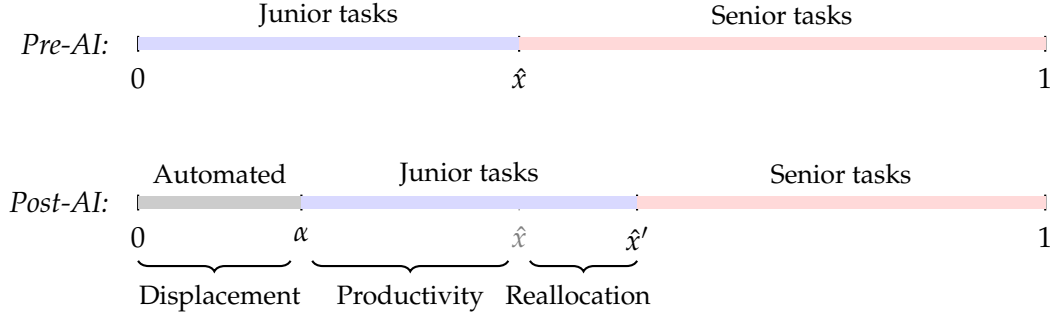


Figure 1: Task Partition Before and After GenAI Adoption

Notes: The figure is schematic and drawn for the maintained case $\alpha < \hat{x}$ (AI automates only junior tasks). The top line shows the pre-AI task partition: juniors perform tasks $x < \hat{x}$ and seniors perform tasks $x \geq \hat{x}$. The bottom line shows the post-AI partition: tasks $x \leq \alpha$ are automated, juniors perform $(\alpha, \hat{x}')$, and seniors perform $[\hat{x}', 1]$. The three components of Proposition 1 correspond to: *displacement*—junior tasks lost to automation $[0, \alpha]$; *productivity*—changes in labor demand on surviving junior tasks $(\alpha, \hat{x})$; and *reallocation*—tasks shifted from seniors to juniors $[\hat{x}, \hat{x}']$.

reflects two channels (see Supplemental Appendix A.6 for the formal derivation). First, a *cost-index channel*: GenAI reduces the CES cost index from P^{pre} to $P^{\text{post}} \leq P^{\text{pre}}$, which lowers labor demand on all surviving tasks regardless of σ . This channel is larger the greater are α (scope of displacement) and δ_J, δ_S (augmentation). Second, a *within-task augmentation channel*, whose sign depends on the elasticity of substitution σ . When $\sigma < 1$, tasks are complementary in production, so the firm requires a roughly fixed mix of task outputs; making juniors more productive per task means fewer are needed, reinforcing the cost-index channel. When $\sigma = 1$ (Cobb-Douglas), only the cost-index channel operates and the productivity effect is still negative. Thus, if $\sigma \leq 1$, the within-task augmentation channel is non-positive and the cost-index channel is negative, making the productivity effect unambiguously negative.¹⁵ When $\sigma > 1$, within-task augmentation tends to raise labor demand and can offset the cost-index channel, making the net productivity effect ambiguous.

Finally, the *reallocation effect* captures juniors' expanded task set due to differential augmentation. The number of tasks reallocated to juniors depends on two forces (see

¹⁵The formal expression, derived in Supplemental Appendix A.6, shows that the ratio of post- to pre-AI labor demand on a surviving task is $\frac{l_J^{\text{post}}(x)}{l_J^{\text{pre}}(x)} = \rho \cdot (1 + \delta_J)^{\sigma-1}$, where $\rho \equiv (P^{\text{post}}/P^{\text{pre}})^{\sigma} \leq 1$ captures the cost-index channel. P is the Lagrange multiplier on the fixed-output constraint—the shadow (unit) cost of output—rather than a market price; since wages are fixed in units of an outside numéraire, P cannot be normalized to one both before and after adoption.

Supplemental Appendix A.6 for the local approximation). It is increasing in the differential augmentation $\delta_J - \delta_S$: the larger the gap in productivity gains across worker types, the more tasks shift from seniors to juniors. It is decreasing in the steepness of comparative advantage at the boundary—i.e., how rapidly the productivity ratio $\gamma_S(x)/\gamma_J(x)$ rises at $\hat{x}$. When this slope is steep, even a large augmentation advantage shifts few tasks. When $\delta_J = \delta_S$, the threshold does not shift and the reallocation effect vanishes entirely.

When tasks are complementary or Cobb-Douglas ($\sigma \leq 1$), two of the three forces reduce junior labor demand: displacement is unambiguously negative and the productivity effect is negative under any GenAI adoption ($\alpha > 0$ or $\delta_J > 0$). The only countervailing force—reallocation—depends on the differential augmentation $\delta_J - \delta_S$ and vanishes entirely when GenAI augments both worker types equally. Junior labor demand therefore declines whenever the reallocation effect is sufficiently small relative to the combined displacement and productivity effects—for example, when comparative advantage is steep near $\hat{x}$ or when the differential augmentation $\delta_J - \delta_S$ is small.

An analogous decomposition holds for seniors. Under the maintained assumption $\alpha < \hat{x}$, no senior tasks are automated, so the displacement effect is zero. The reallocation effect now runs in the opposite direction—seniors lose tasks near $\hat{x}$ to juniors—and the productivity effect operates on surviving senior tasks $[\hat{x}', 1]$. Formally,

$$\Delta L_S^* = \underbrace{0}_{\text{displacement}} + \underbrace{-\int_{\hat{x}}^{\hat{x}'} l_S^{\text{pre}}(x) dx}_{\text{reallocation } (\leq 0)} + \underbrace{\int_{\hat{x}'}^1 [l_S^{\text{post}}(x) - l_S^{\text{pre}}(x)] dx}_{\text{productivity } (\leq 0)}. \quad (6)$$

The reallocation effect is weakly negative: the magnitude of senior task losses depends on $\delta_J - \delta_S$ and the steepness of comparative advantage near $\hat{x}$, and vanishes when $\delta_J = \delta_S$. The productivity effect is non-positive when $\sigma \leq 1$, and smaller in magnitude than for juniors because $\delta_S \leq \delta_J$. Supplemental Appendix A.6 provides the formal derivation.

The two decompositions reveal an asymmetry: juniors uniquely face displacement, while reallocation operates as an offsetting force only for them. Junior employment therefore declines by more than senior employment whenever displacement and productivity effects outweigh reallocation.